

POOJA TALELE

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AI-focused engineer with hands-on experience building production-oriented RAG systems, agentic workflows, and computer vision pipelines. Skilled in optimizing latency, retrieval relevance, and reliability in LLM-powered applications for fast-paced startup environments.

EXPERIENCE

Fanisko
AI Developer Intern

- Built a hybrid RAG-based HR chatbot over **400** policy PDFs, enabling enterprise employee self-service queries.
- Fine-tuned prompts to improve response relevance & domain alignment for HR-specific enterprise queries.
- Optimized chunking, top-k retrieval, prompt optimization, and caching, reducing latency from **5s to 2s**.
- Designed architecture for resume-JD matching and interview automation with relevance scoring.
- Developed a vision pipeline using SAM + inpainting to reconstruct empty shelves from images (**tested on 20-50 images**).
- Presented weekly demos translating research findings into implementable product improvements.

Apr 2025 – Jul 2025
Remote

PROJECTS

Reddit Browser Automation System
Tech: Python, Selenium, Proxy Rotation, Session Persistence

- Built a Reddit automation framework achieving **99%** login success using cookies and session recovery.
- Implemented proxy rotation, retry logic, and network switching to recover automatically from login failures.
- Simulated human-like interaction patterns with randomized scrolling, typing, and delays to reduce bot detection risk.
- Designed modular architecture separating browser, proxy, and action layers for reusable automation workflows and logging.

DSA Teaching Assistant
Tech: React.js, Node.js, Express.js, OpenAI API

- Built a guardrail-first LLM tutor that teaches DSA concepts through hints instead of direct code generation.
- Designed anti-jailbreak prompt logic to block direct-answer extraction and language-based bypass attempts.
- Tested with learners enabling understanding of easy problems within **2-3 prompts** on average.

Blood Report Analysis using CrewAI
Tech: CrewAI, Python, Web Scraping

- Built an agentic workflow to parse medical blood reports and generate post-analysis health recommendations.
- Orchestrated file ingestion, OCR/text extraction, and multi-agent reasoning pipelines using CrewAI.
- Processed reports end-to-end in **7 seconds** with modular agent-based execution.

Person & PPE Detection System
Tech: Python, YOLOv8, OpenCV

- Built a two-stage YOLOv8 pipeline achieving **80%** accuracy by detecting persons first and classifying PPE on cropped regions.
- Processed 1,000 images by converting Pascal VOC annotations, separating labels, and generating YOLO datasets.
- Developed data pipeline to crop person bounding boxes and adjust PPE labels, improving model focus and detection precision.
- Implemented inference pipeline to detect persons and overlay PPE predictions on real-world images for safety use cases.

EDUCATION

Scaler School of Technology
Undergraduate Program in Computer Science

- CGR: **8.2/10**

BITS Pilani
B.Sc (Hons) in Computer Science

- CGPA: **8.4/10**

Jul 2023 – Jul 2027
Bangalore, India

Jul 2023 – Jul 2026
India

TECHNICAL SKILLS

Languages	Python, Java, JavaScript
Backend & Frameworks	React.js, Node.js, Express.js, LangChain
Cloud & DevOps	Docker, Kubernetes, AWS, Git, GitHub
AI & Computer Vision	RAG Systems, Prompt Engineering, CrewAI, YOLOv8, SAM
Databases	MongoDB, MySQL
ML & Data	TensorFlow, Keras, OpenCV
Core Concepts	LLM Workflows, Retrieval Systems, Automation Pipelines
Tools	Postman, VS Code, Jupyter, Google Colab

LEADERSHIP

Scaler School of Technology Student Council
Vice President

- Representing and addressing operational concerns for a student community of **1300+** members.
- Leading an 11-member council focused on designing and improving internal student systems and processes.

2025 – Present
Bangalore, India